

VanniKawachh: A Distributed AI Acoustic Intelligence and Autonomous Drone Response Network for Women Safety

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Abstract—Crimes against women often happen in surveillance and cellular dead zones such as dark streets, forest stretches, and campus outskirts. Most safety technology puts the burden on the victim: panic apps, wearables, and carried devices only work while a charged, reachable device is in the victim’s hand. We present VanniKawachh, a three-tier network that moves the trigger burden from the victim to the infrastructure. Her voice is the trigger. Solar-powered, pole-mounted sensing nodes (ESP32-S3 + INMP441) screen every audio frame on-device with a lightweight MFCC + CNN model (Stage 1, <50 ms budget, recall-tuned), so no continuous audio ever leaves a pole. A Raspberry Pi 5 hub confirms candidate events with PANNs deep audio tagging fused with PIR motion, ambient-light, and time-of-day evidence (Stage 2, precision-tuned), and dispatches only above explicit verification and severity thresholds. A confirmed alert is a 25-byte packet sealed with AES-128-CTR, a truncated HMAC-SHA256 tag, and a per-node replay counter. It carries surveyed node coordinates from a registry rather than a live GPS fix, and travels over LoRa with no cellular dependency to a police dashboard. The same alert auto-dispatches an autonomous quadcopter that records evidence during hover, descends to 3 m, drops a first-aid kit (on failure it returns to launch), and returns. The response layer is our previously SITL-verified flight stack (verified mode transition, a landing-interlocked mission queue, and a debounced severity-ordered failsafe arbiter), carried over unchanged. The complete chain runs end-to-end with zero hardware: the Phase-0 rehearsal passed with fused severity 0.88 and kit release commanded at 3.1 m inside a 328 s SITL mission; the acceptance harness passes 8/8 checks with 0.4 to 0.6 m terminal accuracy; 68 automated unit cases cover the safety logic, packet cryptography, and dispatch gating. Stage-1 model training and hardware range measurements are Phase 1 and 2 work in progress. The development chain used clearly labelled heuristic stand-ins, and this paper reports no acoustic accuracy it has not measured.

Index Terms—women safety; acoustic event detection; TinyML; PANNs; sensor fusion; LoRa; AES-128; autonomous UAV; verified dispatch; SITL.

I. Introduction

Safety technology fails where assaults happen: in places with few cameras, few patrols, and often no cellular coverage, at moments when the victim cannot operate a device because her hands are occupied or the phone is snatched, discharged, or out of reach. The deployed answer to women’s safety has been to instrument the victim with

panic buttons, mobile apps, and smart wearables. The best of these report high trigger accuracy, but their failure mode is structural: a device-borne trigger protects only the moments in which the device is present, charged, and reachable.

VanniKawachh (“voice-shield”) moves the burden. Pole-mounted infrastructure listens. A scream, a cry, or a shouted “help” / “bachao” is the trigger, and the victim needs no device to produce it. The engineering problem then has three parts: (i) detection must run on a solar pole budget without streaming audio anywhere; (ii) false alarms must be suppressed well enough that acting on an alert is defensible, because the action is a drone launch; and (iii) the alert path must work exactly where cellular does not, and it must be unforgeable, because a spoofed alert would launch the drone for an attacker.

Our contributions are:

- 1) An end-to-end architecture covering infrastructure sensing, two-stage verification, offline encrypted alerting, and autonomous field response, where surveyed prior work covers at most one link (Section II).
- 2) A two-stage acoustic pipeline that splits recall and precision between an on-node TinyML screen and a hub-side PANNs + sensor-fusion confirmation, with tested dispatch gating (Section IV).
- 3) A secure LoRa alert protocol: sealed 25-byte packets resolved against a surveyed node registry, so position is never transmitted or trusted from the field (Section V).
- 4) A safety-verified autonomous response layer: our SITL-verified dispatch stack, extended with an evidence camera window and a rule-bounded first-aid kit drop (Section VI).
- 5) A zero-hardware full-chain validation (Phase 0) that runs the integrated system in software-in-the-loop simulation and states which numbers are measured and which await the hardware phases (Section VII).

II. Related Work

Our twelve-paper survey (2023 to 2026; entries [S1] to [S12], full bibliography in the group’s seminar record and

the journal version) [13] splits the field on one question: who carries the trigger?

Victim-carried devices (survey entries [S1], [S5], [S6], [S9], [S10], [S11] of [13]) include apps, wearables, and IoT panic buttons. They report trigger-classification accuracies up to 97.5%, typically with GPS + GSM alerting. All of them fail the same way when the device is absent, damaged, discharged, or unreachable, and they depend on cellular coverage in exactly the locations that lack it.

Detection-only audio systems (entries [S2], [S3], [S4], [S7], [S8] of [13]) reach 92 to 95.5% scream detection with CNN-Transformer hybrids and transfer-learned backbones (InceptionV3, MobileNetV2) on mel-spectrograms. But they run on server-scale compute, are evaluated on curated data, and stop at classification: no location delivery, no dead-zone alerting, no response.

The gap is the chain itself. No surveyed system provides infrastructure sensing, verified offline alerting, and autonomous response end to end. Each link is supported by adjacent work: TensorFlow Lite Micro keyword spotting [7] for the node, PANNs large-scale pretrained audio tagging [6] for the hub, LPWAN practice [8] for the alert path, and the open-source autopilot ecosystem [1], [2] for the response. Our prior flight-stack work supplies the response layer’s safety machinery, including the finding that shaped the design philosophy: flight autopilots can silently reject mode commands while client libraries report success. VanniKawachh applies the resulting rule, a claim is not a confirmation, to every layer: sound, packet, dispatch, and flight mode.

III. System Architecture

Fig. 1 traces one incident through the three tiers: the solar-powered sensing node (ESP32-S3 + INMP441, PIR + LDR context, Stage-1 hit → sealed alert over LoRa + 4 s clip over ESP-NOW/WiFi), the Raspberry Pi 5 hub (unseal, registry lookup, Stage-2 PANNs + fusion, police dashboard, POST /trigger), and the SITL-verified response drone (trigger API → queue → 13-state FSM with hover-record, 3 m kit drop, and RTL).

Three design decisions shape the architecture. Fixed nodes carry no live GPS. Each pole is surveyed once at installation. The hub’s registry maps node_id → (lat, lon), so the radio carries two bytes of identity, and a node has no GPS to spoof, jam, or drain. LoRa carries the alert, never the audio. LoRa’s effective throughput of about 1 to 5.5 kbps cannot move a clip, so the sealed alert goes over LoRa at once while the 4 s verification clip follows over ESP-NOW/WiFi (about 250 kbps, hundreds of metres line of sight). If the clip does not arrive within 8 s, the hub falls back to the Stage-1 confidence scaled by 0.6; it always logs the incident, and dispatches only if the remaining evidence is strong. The flight core is untouched. All v1 safety machinery carries over unchanged, which is why the response half of the system already works.

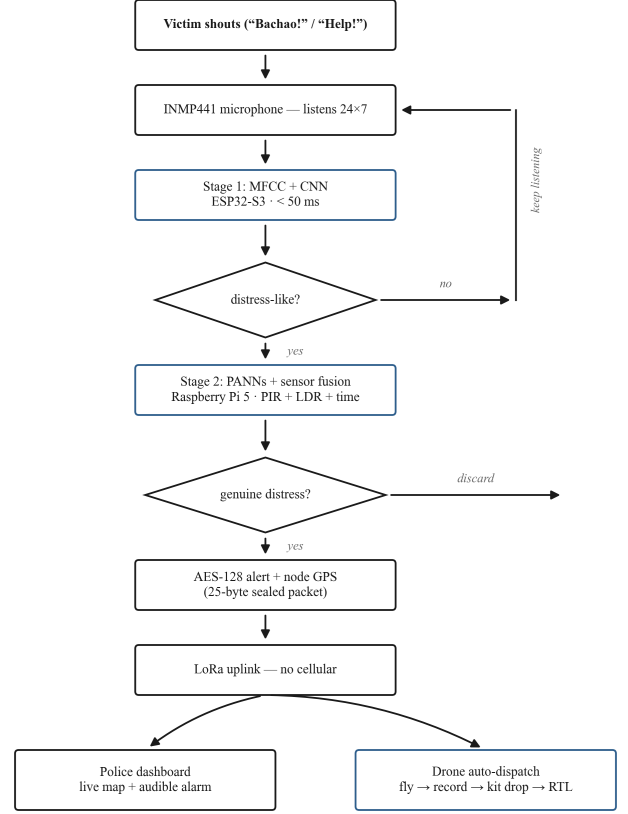


Fig. 1. The VanniKawachh chain, from acoustic event through Stage-1 on-node screening, Stage-2 hub verification and fusion, and sealed LoRa alerting to the parallel dashboard and drone-dispatch paths. Each tier verifies before it acts.

IV. The Two-Stage Acoustic Pipeline

Stage 1 (node, recall-tuned). The ESP32-S3 frames 16 kHz mono audio from the INMP441, extracts MFCCs, and runs a tiny quantized CNN (TensorFlow Lite Micro, micro_speech-class [7]) against the distress vocabulary (scream, cry, and the “help”/“bachao” keywords) within a <50 ms per-frame budget. The front end is the standard speech chain: each frame is pre-emphasised,

$$y[n] = x[n] - 0.97 x[n - 1], \quad (1)$$

Hamming-windowed,

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad (2)$$

and its magnitude spectrum pooled by M triangular filters spaced uniformly on the mel scale

$$m = 2595 \log_{10}\left(1 + \frac{f}{700}\right), \quad (3)$$

giving log filterbank energies

$$e_j = \log \sum_k |X(k)|^2 H_j(k), \quad (4)$$

which a DCT-II decorrelates into the 13 cepstral coefficients used as the per-frame feature vector,

$$c_i = \sum_{j=1}^M e_j \cos \left[i \left(j - \frac{1}{2} \right) \frac{\pi}{M} \right], \quad i = 1, \dots, 13. \quad (5)$$

The CNN classifies each MFCC frame through a softmax output,

$$p_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}, \quad (6)$$

is trained with the cross-entropy loss

$$L = - \sum_i y_i \log p_i, \quad (7)$$

and deploys after int8 post-training quantization,

$$x_q = \text{round} \left(\frac{x}{s} \right) + z, \quad (8)$$

with per-tensor scale s and zero-point z . This quantization step fits the model into TFLM on the ESP32-S3. Frames that do not trip Stage 1 are discarded on the spot: nothing is stored and nothing is transmitted. Stage 1 exists to avoid misses; its false positives are expected and cheap because Stage 2 filters them. (The model itself is a hook in the current firmware; training and flashing it is Phase-1 work, see Section VII-C.)

Stage 2 (hub, precision-tuned). The hub re-scores the clip with PANNs [6], the pretrained AudioSet tagging network (CNN14, or a lighter checkpoint on a slow Pi). It takes the summed probability over the distress-relevant AudioSet classes (screaming, shouting, yelling, crying, wailing, and similar) as a distress score in $[0, 1]$. No bespoke training is required; the hub relies on AudioSet scale, which a locality Pi 5 can afford and a solar pole cannot. A labelled energy-heuristic fallback backend (loud, high spectral centroid, bursty) exists so the whole chain runs on any development machine. It is not a claim of accuracy, and every result produced with it is marked as fallback.

Fusion. A night-time scream in a dark spot with motion nearby carries more weight than a daytime shout on a busy road. With a the Stage-2 audio score, c the Stage-1 confidence, $p \in \{0, 1\}$ the PIR motion flag, $d = 1 - L/255$ the darkness derived from the LDR level $L \in [0, 255]$, and $n \in \{0, 1\}$ the night indicator (1 during 20:00 to 06:00), the fused severity is

$$S = 0.60a + 0.15c + 0.10p + 0.08d + 0.07n, \quad (9)$$

with priority high at $S \geq 0.75$ or verified audio ($a \geq 0.6$) together with PIR motion ($p = 1$). Dispatch requires both gates to pass,

$$\text{dispatch} \iff (a \geq \tau_v) \wedge (S \geq \tau_d), \quad \tau_v = 0.50, \tau_d = 0.60; \quad (10)$$

below either threshold the incident is logged with a human-readable reasons trace and no drone flies. When

Scaled alert packet — 25 bytes total (one LoRa frame)

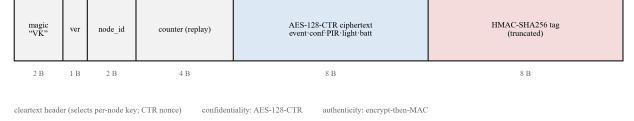


Fig. 2. The 25-byte sealed alert packet: cleartext header, AES-128-CTR ciphertext, and truncated HMAC-SHA256 tag, all in one LoRa frame.

the verification clip never arrives (Section III), the degraded audio score

$$a = 0.6c \quad (11)$$

is substituted before applying (9) and (10). The gating is covered by automated tests, and the weights in (9) are prototype values to be tuned against Phase-1 bench data.

V. Secure LoRa Alerting

A spoofed packet would launch a drone; a replayed one would launch it at a time the attacker chooses. Every alert is therefore a sealed 25-byte packet that fits in one LoRa frame, laid out as shown in Fig. 2: a cleartext header (magic, version, node_id uint16, counter uint32), an 8-byte AES-128-CTR-encrypted payload (event class, Stage-1 confidence, PIR flag, LDR level, node battery), and an 8-byte MAC (HMAC-SHA256 over header + ciphertext, truncated).

Per-node keys derive from the master key K_m by keyed hashing (FIPS-197 AES, RFC 2104 HMAC [10]),

$$K_n = \text{HMAC-SHA256}(K_m, \text{"node:"} \parallel n) [0:16], \quad (12)$$

so provisioning needs only the master key and an id. A captured node compromises one pole, and that node can be revoked in the registry. The payload P is encrypted in CTR mode,

$$C = P \oplus E_{K_n}(\text{IV}), \quad (13)$$

with the counter block IV built from the cleartext header (magic $\parallel$ ver $\parallel$ node_id $\parallel$ counter), which is unique per packet while the counter is monotonic. Authentication is encrypt-then-MAC,

$$\tau = \text{HMAC-SHA256}(K_n, \text{header} \parallel C) [0:8]. \quad (14)$$

The hub accepts a packet only when

$$\tau \text{ valid} \wedge \text{counter} > \text{counter}_{\text{last}}, \quad (15)$$

so bad MACs, unknown node ids, and replays are all rejected. Forging the 64-bit truncated tag succeeds with probability 2^{-64} per attempt. Above the radio, the hub reaches the drone stack through the same token-authenticated, geofence-validated API as any operator; there is no privileged backdoor. Residual risk is stated: jamming can deny service but cannot forge an alert;

multi-node corroboration and node-liveness monitoring are future answers.

The LoRa/WiFi split of Section III follows from the physics. LoRa's range comes from spreading. Each symbol lasts

$$T_s = \frac{2^{\text{SF}}}{\text{BW}}, \quad (16)$$

and the receiver sensitivity floor

$$S_{\text{dBm}} = -174 + 10 \log_{10}(\text{BW}) + \text{NF} + \text{SNR}_{\text{min}} \quad (17)$$

drops below -120 dBm at high SF over $\text{BW} = 125$ kHz, so milliwatts reach kilometres. The same trade caps throughput: the standard LoRa time-on-air expression (Semtech SX127x datasheet [8]) gives ≈ 0.21 s for the 25-byte sealed alert at SF9, BW 125 kHz, CR 4/5, explicit header + CRC, whereas the 4 s clip (≈ 128 kB at ≈ 5.5 kb/s effective) would occupy the channel for minutes. So the alert goes over LoRa and the clip over ESP-NOW/WiFi.

Privacy is built into the design: continuous audio cannot leave a node because the transport cannot carry it and the clip path is event-gated in firmware. Only event-triggered clips ≤ 5 s are ever transmitted, and the alert itself is encrypted.

VI. Safety-Verified Drone Response

The response layer is our SITL-verified dispatch stack. Its three mechanisms transfer intact. Verified mode transition: every flight-mode command, nominal or emergency, routes through a single routine. The routine re-issues the request through layered MAVLink encodings (COMMAND_LONG/MAV_CMD_DO_SET_MODE plus the legacy SET_MODE) every 700 ms until the autopilot's own HEARTBEAT-derived mode confirms adoption, with a cross-action fallback ($\text{RTL} \rightleftharpoons \text{LAND}$) on the abort path. This design answers the silent rejection we observed on ArduCopter 3.3 SITL. Landing interlock: every abnormal termination blocks until the vehicle demonstrably lands and disarms (bounded 240 s) before the queue regains control, and a dequeued mission refuses to arm an armed vehicle. Together, these mean the queue can never start a flight against an airborne vehicle. Failsafe arbitration: battery, GPS, geofence, and timeout monitors feed a 1 Hz arbiter with monotone severity (LAND never downgraded), N -sample GPS debounce, fire-once event semantics, and mid-RTL escalation. Geofence radii, arrival detection against the 5 m waypoint tolerance, and ETA all use the haversine great-circle distance

$$d = 2R \arcsin \sqrt{\sin^2 \frac{\Delta\varphi}{2} + \cos \varphi_1 \cos \varphi_2 \sin^2 \frac{\Delta\lambda}{2}}, \quad (18)$$

$R = 6371$ km.

The thirteen-state mission FSM within which these mechanisms run is shown in Fig. 3.

v2 adds the response payload. The mission FSM (now thirteen states) records camera evidence during the

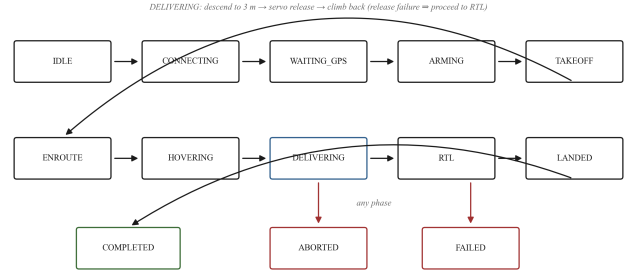


Fig. 3. The 13-state mission FSM, from IDLE through TAKE-OFF, ENROUTE, HOVERING, and DELIVERING to RTL and LANDED, with the abnormal paths into ABORTED and FAILED.

hover window (Pi Camera Module 3 on hardware; a no-op stub in SITL, so the flow is byte-identical) and then enters DELIVERING: descend over the incident point to the configured drop altitude (3.0 m, tolerance 0.7 m, bounded at 45 s), release the first-aid kit via an SG90 servo on Pixhawk AUX OUT 1 commanded with MAV_CMD_DO_SET_SERVO (open 1900 PWM, 2 s settle, re-close 1100), then climb back to cruise altitude for the RTL. The 3 m rule comes from simple ballistics: a kit released from height h free-falls for

$$t = \sqrt{\frac{2h}{g}} \approx 0.78 \text{ s} \quad \text{at } h = 3 \text{ m}, \quad (19)$$

during which a $v_w = 2$ m/s wind drifts it only

$$x \approx v_w t \approx 1.6 \text{ m}, \quad (20)$$

keeping the kit within reach of the victim. A cruise-altitude release would increase both drift and impact energy. The governing rule is that a failed release is never a reason to loiter: the failure is logged and the drone proceeds to RTL regardless. The DELIVERING loop polls the failsafe arbiter like every other phase, so a battery or GPS demand pre-empt the drop. All prototype flying is VLOS-only on a registered airframe with an RC safety pilot, per India's Drone Rules 2021 [5]; autonomous BVLOS response is described strictly as a supervised pilot-program pathway.

VII. Evaluation

A. Methodology

Validation is staged. Tier 1: 68 automated unit cases: 47 over the flight stack's safety logic (every arbiter rule, queue semantics, persistence, every edge-validation bound), 14 over the hub chain (packet seal/unseal, MAC tamper rejection, replay rejection, registry, fusion, pipeline gating, where no dispatch below threshold is asserted rather than assumed, and dispatcher payload shape), and 7 over obstacle keep-out routing. The suites discriminate: run against the pre-hardening implementation, the debounce, cancel, and interlock cases fail. Tier 2: an eight-property

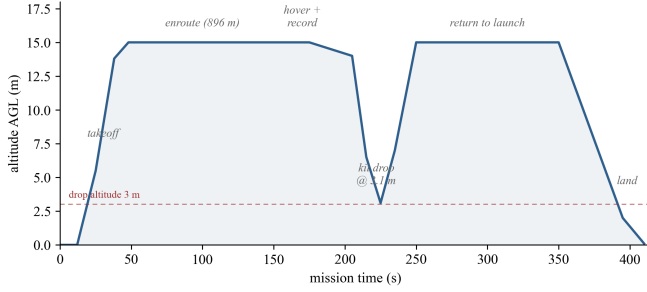


Fig. 4. Altitude profile of the Phase-0 full-chain rehearsal: takeoff to cruise, transit, hover-observe, descent to the 3.1 m kit release, climb-out, and return to launch, completing in 328 s.

SITL acceptance flight (ArduCopter 3.3; 896 m mission at 15 m altitude): simulator up, API up, connected, armed, took off, reached target (≤ 5 m), returned home (≤ 10 m, required), landed. Tier 3: the Phase-0 full-chain rehearsal (scripts/demo_phase0.py). A synthesized distress WAV stands in for the microphone; a simulated node 600 m from home seals a real packet with the production cryptography; the production hub pipeline verifies (fallback backend), fuses, gates, and dispatches; and the production drone stack flies the SITL mission with hover-record and kit drop. Everything except the audio source, the Stage-2 backend, and the physics is production code.

B. Results

Tier 1: 68/68 pass. Tier 2: 8/8 checks pass. Across six acceptance missions spanning the development arc, closest approach was 0.4 to 0.6 m against the 5 m tolerance, and final distance from home was 0.0 to 0.2 m; terminal accuracy is bounded by the autopilot’s loiter behaviour, not by the dispatch layer. Tier 3: the rehearsal passed end to end. The sealed alert (event scream, PIR active, dark LDR) was authenticated and replay-checked; the fallback backend scored the clip above the 0.50 verification threshold; fused severity 0.88 (priority high) cleared the 0.60 dispatch gate; and the SITL mission ran the full lifecycle with the recording window active and kit release commanded at 3.1 m relative altitude, completing in 328 s with 0.4 m closest approach. Fig. 4 shows the rehearsal’s altitude profile, with the descent to the 3.1 m release point visible between the hover window and the return leg.

C. What Is Measured vs. In Progress

The Phase-0 result proves the architecture and the integration: every interface a hardware phase will use (packet bytes, clip convention, thresholds, trigger payload, servo command) was exercised in production form. It deliberately proves nothing about acoustic detection performance. No Stage-1 accuracy exists. The TFLM model is a hook; training and flashing it is Phase 1; <50 ms is a design budget, not a measurement. No field

Stage-2 accuracy exists. The Phase-0 score came from the labelled heuristic fallback on synthesized audio; PANNs’ published performance [6] motivates the backend and is not a field claim. For transparency, the fallback scorer is fully specified: over the clip samples it computes the RMS level

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_n x^2[n]}, \quad (21)$$

the spectral centroid

$$C = \frac{\sum_f f |X(f)|}{\sum_f |X(f)|}, \quad (22)$$

and an envelope-burstiness term, combined as

$$\begin{aligned} \text{score} = & 0.45 \min\left(1, \frac{\text{RMS}}{0.15}\right) \\ & + 0.35 \text{clip}\left(\frac{C - 400}{1600}, 0, 1\right) + 0.20 \text{burstiness}. \end{aligned} \quad (23)$$

Equations (21) to (23) describe a labelled development stand-in (loud, high-centroid, bursty audio scores high), not a detection claim. No radio range/loss or ESP-NOW reliability figures exist (Phase 2). No hardware flight has occurred. Phase 3’s staged VLOS progression governs the transition; the SITL firmware is the 2015-vintage 3.3 build, whose command-delivery faults forced the defensive design. These measurements (outdoor detection distance vs. SNR, per-stage latency, end-to-end false-positive rate on street noise, LoRa range vs. spreading factor) are the explicit deliverables of the Phase 1 and 2 bench campaigns and will appear in the journal version once measured.

VIII. Regulatory, Spectrum, and Privacy Grounding

Deployment legality is part of the design. Airspace: UAS operation in India is governed by The Drone Rules, 2021 (G.S.R. 589(E), Ministry of Civil Aviation) [5], which require registration on the DGCA’s Digital Sky platform and divide airspace into green, yellow, and red zones. All prototype flying sits in the least-restrictive cell of that framework: a registered airframe, VLOS-only, in green-zone private airspace, with an RC safety pilot holding override authority. Autonomous BVLOS response is framed strictly as a supervised pilot-program pathway contingent on regulatory approval, which the Rules and the DGCA BVLOS experiment schemes provide. Spectrum: the prototype’s SX1278 links operate as low-power 433 MHz short-range devices; production would migrate to the WPC-delicensed 865 to 867 MHz band [11], where higher power is permitted licence-free and Indian LoRa deployments normally operate. Neither band needs an operating licence at the power levels used. Privacy: the Digital Personal Data Protection Act, 2023 [12] requires purpose limitation, minimisation, and safeguards. The design answers structurally: on-device processing, no continuous recording or transmission, only event-triggered

clips ≤ 5 s retained as incident evidence, and encryption in transit. Compliance is then a process on top of an architecture that already collects the minimum.

IX. Conclusion

VanniKawachh integrates TinyML pole-side screening, pretrained deep audio verification with environmental fusion, sealed operator-free alerting, and a safety-interlocked autonomous first response into a single women-safety chain that asks nothing of the victim but her voice. The complete chain is demonstrated in simulation before any hardware is committed. One rule governs every layer: a claim is not a confirmation. A Stage-1 hit is not an incident until Stage 2 and fusion say so; a packet is not an alert until its MAC and counter say so; a mode command is not a mode until the autopilot’s telemetry says so; a mission is not finished until the vehicle is disarmed on the ground. Under that rule the integrated rehearsal passed on the first architecture, and the remaining work is measurement and hardening rather than redesign. Future work: the Phase 1 and 2 measurement campaigns; live RTSP/WebRTC streaming to police; OpenCV victim tracking during hover; TDOA multi-node localization between poles; and the city-scale node mesh with fleet dispatch.

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Reproducibility and Originality Statement

The full chain is reproducible on any machine with no hardware and no paid service: git clone <https://github.com/SV-1411/drone.git>, a Python 3.10+ environment, pip install -r requirements-dev.txt plus requirements-hub.txt, then python -m pytest (tier 1), python tests/test_full_mission.py (tier 2, boots SITL locally), and python scripts/demo_phase0.py (tier 3, the full-chain rehearsal). All prose was written for this work; protocol, parameter, and flight-mode names are protocol- or firmware-defined identifiers. A large-language-model assistant was used during drafting for structure and consistency; all technical claims derive from the implementation and test logs in the repository, and the final text was reviewed and accepted by the authors.

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